

Cybersecurity Analysis and Mitigation Plan for Xfibility Bank

Presented By Hashan Kodippilige

Introduction

Cybersecurity attacks are a major risk for banking industry. This investigation aims to evaluate banking webserver network activity and potential security breach using network and process monitoring tools.

After evaluating the results of investigation will proceed to introduce necessary security control measures and policy to safeguard the XFibility bank network.

Abbreviations

- FTP - File Transfer Protocol
- SFTP - Secure File Transfer Protocol
- TCP - Transmission Control Protocol
- IP - Internet Protocol
- VM - Virtual Machine
- IDS - Intrusion Detection System
- DLP - Data Loss Prevention
- NIST - National Institute of Standards and Technology

Goal

To use different tools to analyze data from a cybersecurity incident, reconstruct and correlate events, and communicate findings.

Tools used for the Investigation

- Wireshark - Packet analysis tool used to capture and inspect network traffic.
- Network Miner - Network forensic analysis tool that extracts hosts, credentials, and transferred files from packet captures
- Process Monitor - Tracks real-time system operations such as file access, registry changes, and network connections

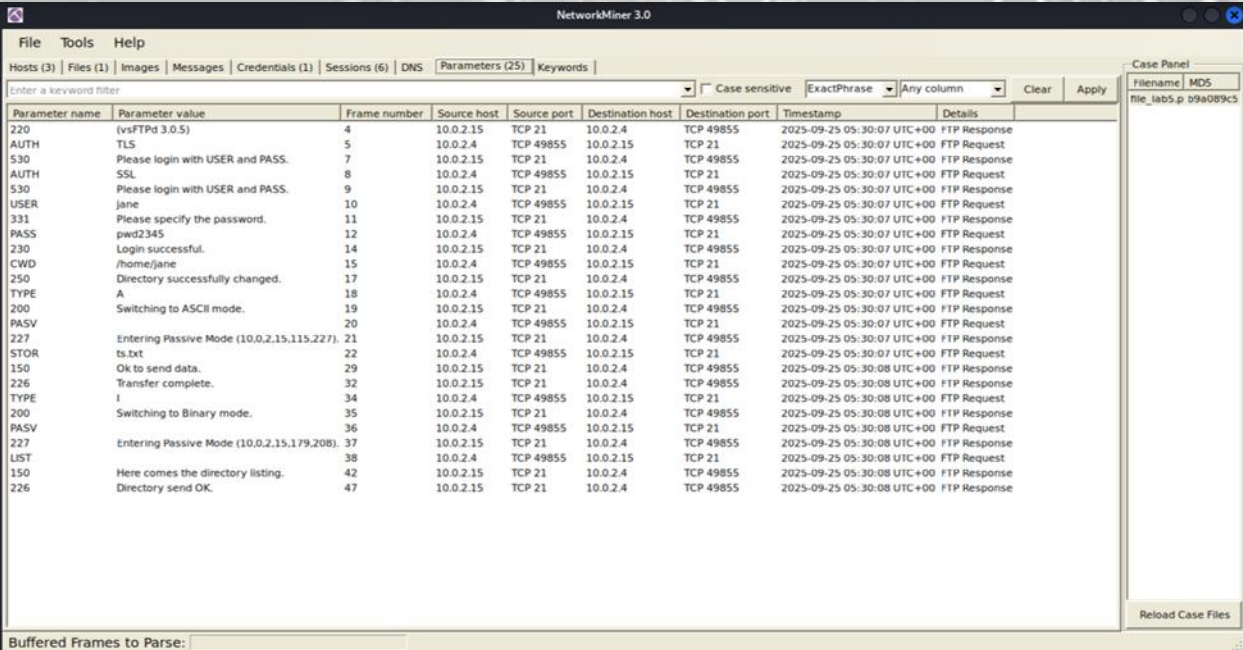
From Wireshark

- TCP and FTP protocols were used to reach the bank IP (Victim IP)
- Shown the successful login and file upload through FTP
- Ports involved: 21 (FTP) and ephemeral ports 49855 and 49857

FileEditViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

From Network Miner

- FTP commands including USER jane, PASS pwd2345
- Directory changed to /home/jane
- Transfer file : ts.txt (file transfer completion code 226)



NetworkMiner 3.0

File Tools Help

Hosts (3) | Files (1) | Images | Messages | Credentials (1) | Sessions (6) | DNS | Parameters (25) | Keywords

Enter a keyword filter

Case sensitive ExactPhrase Any column Clear Apply

Parameter name	Parameter value	Frame number	Source host	Source port	Destination host	Destination port	Timestamp	Details
220	(vsFTPD 3.0.5)	4	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
AUTH	TLS	5	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
530	Please login with USER and PASS.	7	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
AUTH	SSL	8	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
530	Please login with USER and PASS.	9	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
USER	jane	10	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
331	Please specify the password.	11	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
PASS	pwd2345	12	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
230	Login successful.	14	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
CWD	/home/jane	15	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
250	Directory successfully changed.	17	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
TYPE	A	18	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
200	Switching to ASCII mode.	19	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
PASV		20	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
227	Entering Passive Mode (10.0.2.15,115,227)	21	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
STOR	ts.txt	22	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
150	Ok to send data.	29	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:08 UTC+00	FTP Response
226	Transfer complete.	32	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:08 UTC+00	FTP Response
TYPE	I	34	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:08 UTC+00	FTP Request
200	Switching to Binary mode.	35	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:08 UTC+00	FTP Response
PASV		36	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:08 UTC+00	FTP Request
227	Entering Passive Mode (10.0.2.15,179,208)	37	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:08 UTC+00	FTP Response
LIST		38	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:08 UTC+00	FTP Request
150	Here comes the directory listing.	42	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:08 UTC+00	FTP Response
226	Directory send OK.	47	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:08 UTC+00	FTP Response

Case Panel

Filename MD5

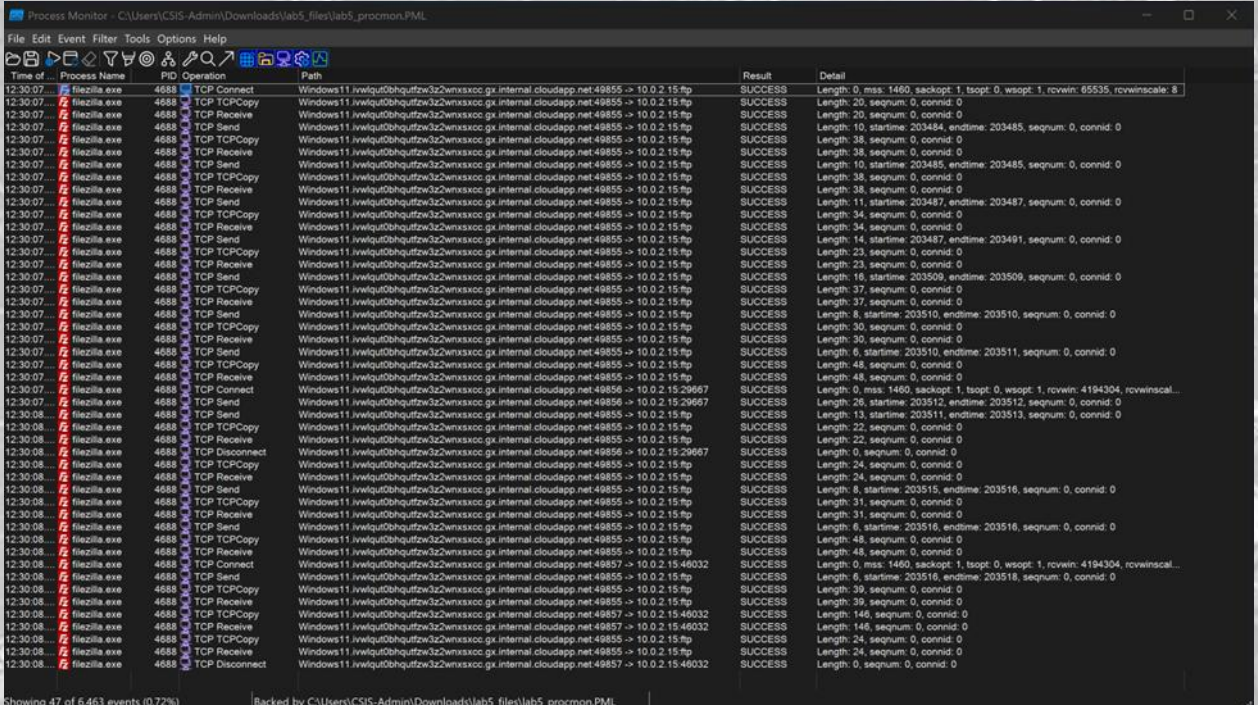
file_lab5.p b9a089c5

Reload Case Files

Buffered Frames to Parse:

From Process Monitor

- Obtained details of FileZilla TCP Connect, TCP Receive, TCP Send, and TCP Disconnect events to 10.0.2.15 and data ports 29667, 46032



The screenshot shows the Process Monitor application window with the following details:

- Title Bar:** Process Monitor - C:\Users\CSIS-Admin\Downloads\lab5_files\lab5_prcmon.PML
- Menu Bar:** File Edit Event Filter Tools Options Help
- Toolbar:** Includes icons for file operations, filters, and search.
- Table Columns:** Time of..., Process Name, PID, Operation, Path, Result, Detail.
- Table Content:** A list of network events for FileZilla.exe (PID 4688) involving 10.0.2.15. The events include TCP Connect, TCP Receive, TCP Send, TCP Copy, and TCP Disconnect. The 'Detail' column provides specific network parameters such as sequence numbers, window sizes, and flags.
- Status Bar:** Shows 'Showing 47 of 6,463 events (0.72%)' and 'Backed by C:\Users\CSIS-Admin\Downloads\lab5_files\lab5_prcmon.PML'.

Findings

- A suspicious FTP connection detected between Victim IP 10.0.2.4 and Attacker IP 10.0.2.15
- Attack occurred over insecure FTP communication
- Unencrypted credentials: USER jane, PASS pwd2345
- File transfer confirmed: ts.txt (file transfer completion code 226)

Recommended Controls

- Replace FTP with secure File transfer protocols (SFTP)
- Implement Multi Factor Authentication (MFA)
- Implement Network Segmentation and Firewalls
- Use Data Loss Prevention (DLP) Tools
- Deploy real-time Endpoint Detection and Response (EDR)
- Implement Policies to mitigate Cyber Incidents
- Conduct security awareness programs periodically

Conclusion

- Network Tools confirmed successful data exfiltration
- FTP vulnerability exploited due to weak encryption
- Can be reduced risk by implementing recommended controls
- Single user mistake will provide attacker to successful data exfiltration from banking network
- Data theft will lead bank to financial loss, and reputation damage

References

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THANK YOU!